

```

import numpy as np
import pandas as pd

raw_data={
    "student":["Ankush","Sahil","Abhimanyu","Riya"],
    "Maths":[85,78,92,88],
    "Stats":[90,82,79,95],
    "DSA":[88,91,85,80],
}

df=pd.DataFrame(raw_data)
print("=" * 55)
print("STEP 1: Raw Data converted to DataFrame")
print("=" * 55)
print("=" * 55)
print(df)
print()

marks_array = df[["Maths", "Stats", "DSA"]].to_numpy()
print("=" * 55)
print("STEP 2: NumPy Array Properties")
print("=" * 55)
print("Array:\n", marks_array)
print("Shape:", marks_array.shape)
print("Size:", marks_array.size)
print("Dimension:",marks_array.ndim)
print()

A = np.array([[3, 2, 1],
              [1, 1, 2],
              [2, 1, 3]])

B = np.array([[1, 1, 1],
              [2, 2, 2],
              [3, 3, 3]])

print("=" * 55)
print("STEP 3: Matrix Operations")
print("=" * 55)
print("Matrix A:\n", A)
print("\nMatrix B:\n", B)

print("\n--- Addition (A + B) ---")
print(A + B)

print("\n--- Subtraction (A - B) ---")
print(A - B)

print("\n--- Element-wise Multiplication (A * B) ---")
print(A * B)

print("\n--- Matrix Product (A @ B) ---")
print(A @ B) # same as np.dot(A, B) or np.matmul(A, B)

print("\n--- Transpose of A (A.T) ---")
print(A.T)

print("\n--- Inverse of A (np.linalg.inv) ---")

try:
    A_inv = np.linalg.inv(A)
    print(A_inv)
except np.linalg.LinAlgError:
    print("Matrix A is singular – inverse does not exist.")

```

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=====
STEP 1: Raw Data converted to DataFrame
=====
=====
      student  Maths  Stats  DSA
0      Ankush    85    90   88
1       Sahil    78    82   91
2  Abhimanyu    92    79   85
3        Riya    88    95   80

=====
STEP 2: NumPy Array Properties
=====
Array:
[[85 90 88]
 [78 82 91]
 [92 79 85]
 [88 95 80]]
Shape: (4, 3)
Size: 12
Dimension: 2

=====
STEP 3: Matrix Operations
=====
Matrix A:
[[3 2 1]
 [1 1 2]
 [2 1 3]]

Matrix B:
[[1 1 1]
 [2 2 2]
 [3 3 3]]

--- Addition (A + B) ---
[[4 3 2]
 [3 3 4]
 [5 4 6]]

--- Subtraction (A - B) ---
[[ 2  1  0]
 [-1 -1  0]
 [-1 -2  0]]

--- Element-wise Multiplication (A * B) ---
[[3 2 1]
 [2 2 4]
 [6 3 9]]

--- Matrix Product (A @ B) ---
[[10 10 10]
 [ 9  9  9]
 [13 13 13]]

--- Transpose of A (A.T) ---
[[3 1 2]
 [2 1 1]]

```

